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Inventor(s)	Suzuki; Michiyasu et al.

Subdural sensor

Abstract

A subdural sensor includes: a substrate formed of a flexible material; and at least one type of sensor part mounted on the substrate. The substrate has an elongated shape, and includes: a sensor area in which the sensor part is mounted and a wiring pattern connected to the sensor part is formed; a wiring area contiguous with the sensor area, the wiring pattern extending in the wiring area; and a connector area contiguous with the wiring area, the connector area being an area on which a connector to be connected to the wiring pattern extending from the wiring area is mounted. A tip part of the sensor area has a planar shape that curves convexly toward an outer periphery, and a side shape that curves toward a first surface, the first surface being on the side of a dura mater when the subdural sensor is inserted into the subdural space.

Inventors: Suzuki; Michiyasu (Yamaguchi, JP), Nomura; Sadahiro (Yamaguchi, JP), Inoue; Takao (Yamaguchi, JP), Yamakawa; Toshitaka (Kumamoto, JP)

Applicant: ANT5 Co., Ltd. (Yamaguchi, JP)

Family ID: 1000008782978

Assignee: ANT5 Co., Ltd. (Yamaguchi, JP)

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Primary Examiner: Taylor, Jr.; Duane N

Attorney, Agent or Firm: Nakanishi IP Associates, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application of PCT International Application No. PCT/JP2021/024617 filed on Jun. 29, 2021, which designated the United States, and which claims the benefit of priority from Japanese Patent Application No. 2020-112158, filed on Jun. 29, 2020. The entire contents of these applications are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to subdural sensors for acquiring biological information by being arranged in a subdural space.

Description of the Related Art

(2) In the treatment of patients with brain disease or brain injury, in addition to identifying the symptoms themselves, such as increase in intracranial pressure (ICP), epilepsy, and reduction in cerebral blood flow, it is important to accurately monitor brain pathological conditions ahead of the appearance of such symptoms and/or the occurrence of severe events in order to identify risks more quickly.

(3) Regarding the acquisition of biological information of the brain, for example, JP6296606 B and Toshitaka Yamakawa, et al., “Implantable Multi-Modality Probe for Subdural Simultaneous Measurement of Electrophysiology, Hemodynamics, and Temperature Distribution”, IEEE Transactions on Biomedical Engineering, Vol. 66, No. 11, November 2019, pp. 3204-3211, disclose subdural sensors that make contact with or are inserted into the subdural space and measure hemodynamics at least close to the brain surface. The subdural sensors disclosed in JP6296606 B and Yamakawa, et al. can simultaneously measure cerebral blood flow, brain tissue oxygen saturation (brain StO₂), electroencephalogram, and brain temperature.

(4) Edema or hematoma resulting from cerebral infarction or trauma may result in increased intracranial pressure (ICP), which may cause brain hypoxia or neuronal death due to the reduction of cerebral blood flow. Decompressive craniotomy may be performed to reduce ICP. Another serious sequela is elevation of brain temperature causing neuronal death. In order to suppress this temperature elevation, cerebral hypothermia therapy is established, which cools the brain to a therapeutic temperature level predetermined. As an example, JP2011-083315 A discloses a system for regionally cooling the brain. Although the cerebral protective effect by general hypothermia by which the whole body is cooled, little is known about focal brain cooling. Therefore, sufficient brain information obtained from this sensor is quite important in order to exert cerebral protective effects by the focal cooling.

BRIEF SUMMARY OF THE INVENTION

(5) A subdural sensor, which is one aspect of the present invention, is to be arranged in a subdural space and acquires biological information about the brain. The subdural sensor comprises: a substrate formed of a flexible material; and at least one type of sensor part mounted on the substrate, wherein the substrate has an elongated shape as a whole, wherein the substrate includes: a sensor area in which the at least one type of sensor part is mounted and a wiring pattern connected to the at least one type of sensor part is formed; a wiring area contiguous with the sensor area on one end thereof, the wiring pattern extending in the wiring area; and a connector area contiguous with the other end of the wiring area, the connector area being an area on which a connector to be connected to the wiring pattern extending from the wiring area is mounted, and a tip part of the sensor area has a planar shape that curves convexly toward an outer periphery, and a side shape that curves toward a first surface, the first surface being on the side of a dura mater when the subdural sensor is inserted into the subdural space.

(6) The above-described and other features, advantages and technical and industrial significance of the present invention, will be better understood by reading the following detailed description of the current preferred embodiments of the present invention while considering the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a plan view schematically showing a portion of a subdural sensor according to one embodiment of the present invention.
- (2) FIG. 1B is a side view schematically showing the portion of the subdural sensor according to one embodiment of the present invention.
- (3) FIG. 2 is a plan view schematically showing a portion of a subdural sensor according to one embodiment of the present invention.
- (4) FIG. 3 is a partial cross-sectional view showing the vicinity of a light reflection part shown in FIG. 1A.
- (5) FIG. 4 is a schematic diagram showing a usage example of a subdural sensor according to one embodiment of the present invention.
- (6) FIG. 5A is a plan view schematically showing a first variation of a tip part of a substrate.
- (7) FIG. 5B is a side view schematically showing the first variation of the tip part of the substrate.
- (8) FIG. 6 is a plan view schematically showing a second variation of the tip part of the substrate.
- (9) FIG. 7 is a plan view schematically showing a first variation of a wiring area of the substrate.
- (10) FIG. 8 is a plan view schematically showing a second variation of the wiring area of the substrate.
- (11) FIG. 9A is a plan view schematically showing a third variation of the wiring area (on the sensor area side) of the substrate.
- (12) FIG. 9B is a plan view schematically showing the third variation of the wiring area (on the connector area side) of the substrate.
- (13) FIG. 9C is a plan view schematically showing the third variation of the wiring area (on the connector area side) of the substrate.
- (14) FIG. 10A is a plan view (seen from the surface) schematically showing an implementation of an intracranial pressure sensor.
- (15) FIG. 10B is a side view schematically showing the implementation of the intracranial pressure sensor.
- (16) FIG. 10C is a plan view (seen from the back) schematically showing the implementation of the intracranial pressure sensor.
- (17) FIG. 11 is a plan view (seen from the back) schematically showing another implementation of the intracranial pressure sensor.
- (18) FIG. 12 is a partial cross-sectional view showing a variation of the light reflection part.
- (19) FIG. 13 is a partial cross-sectional view of another variation of the light reflection part.
- (20) FIG. 14 is a plan view schematically showing a variation of the substrate shape.
- (21) FIG. 15 is a plan view schematically showing another variation of the substrate shape.
- (22) FIG. 16 is a plan view illustrating the wiring in a connector area.
- (23) FIG. 17 is a plan view showing a variation of the wiring pattern in the wiring area.
- (24) FIG. 18 is a partial cross-sectional view for describing an implementation of an element.
- (25) FIG. 19 is a plan view schematically showing a further variation of the subdural sensor.
- (26) FIG. 20 is a plan view schematically showing a further variation of the subdural sensor.

DETAILED DESCRIPTION OF THE INVENTION

(27) Hereinafter, a subdural sensor according to embodiments of the present invention will be described with reference to the drawings. It should be noted that the present invention is not limited

by these embodiments. In the description of each drawing, the same parts are denoted by the same reference numbers.

(28) The drawings referred to in the following description are merely schematic representations of shape, size, and positional relationship to the extent that the subject matter of the present invention may be understood. In other words, the present invention is not limited only to the shapes, sizes, and positional relationships illustrated in the respective figures. In addition, the drawings may also include, among themselves, parts having different dimensional relationships and ratios from each other.

Embodiments of Subdural Sensor

(29) FIG. 1A is a plan view schematically showing a portion of a subdural sensor according to one embodiment of the present invention. FIG. 1B is a side view schematically showing the portion of the same subdural sensor. FIG. 2 is a plan view schematically showing another portion of the same subdural sensor. The subdural sensor **10** according to the present embodiment is a device arranged in the subdural space and acquires biological information about the brain. The sensor is equipped with a substrate **100** formed of a flexible material and at least one type of sensor part mounted on the substrate **100**.

(30) The substrate **100** is a so-called flexible substrate that is flexible and is formed of resin materials, such as polyimide. The substrate **100** and the sensor parts are coated in an integral manner with biocompatible materials such as Parylene (registered trademark), except for the vicinity of the apex of a light reflection part **113**, which will be described later. The thickness of the film **107** is not particularly limited, but in the case of using, for example, Parylene (registered trademark), the thickness may preferably be set to, for example, 5 μm to 20 μm , in order to prevent significant loss of the flexibility of the substrate **100**.

(31) As shown in FIGS. 1A to 2, the substrate **100** has a generally elongate shape, and includes a sensor area **101** in which at least one type of sensor part is arranged, a wiring area **102** that is continuous with the base end of the sensor area **101** on one end thereof, and a connector area **103** that is continuous with the other end of the wiring area **102**. These sensor area **101**, wiring area **102**, and connector area **103** are preferably formed in an integral manner by a single flexible substrate. In the sensor area **101**, the wiring area **102**, and the connector area **103**, a wiring pattern connected to the sensor parts is continuously formed. The wiring pattern may be formed only on one surface or on both surfaces. Alternatively, it may be formed over several layers.

(32) The sensor area **101** is an area inserted into the subdural space, where the sensor parts are mounted and the wiring pattern connected to the sensor parts is formed. Here, the sensor area **101** is inserted in the subdural space so that the surface (first surface) **104** is on the side of the dura mater and the back (second surface) **105** is on the side of the brain surface. FIG. 1A shows a plurality of sensor parts arranged in a line along the longitudinal direction on the back **105** of the sensor area **101**.

(33) Preferably, the tip part **10a** of the sensor area **101** has a planar shape that curves convexly toward the outer periphery, as shown in FIG. 1A. In addition, the tip part **10a** preferably has a side shape that curves toward the surface **104**, as shown in FIG. 1B.

(34) A portion of the wiring area **102** on the side of the sensor area **101** is placed in vivo together with the sensor area **101**. On the other hand, a portion of the wiring area **102** on the side of the connector area **103** is arranged in vitro together with the connector area **103**. The wiring area **102** and the sensor area **101** are preferably connected at an obtuse angle or in a gentle curve in order to prevent disconnection of the substrate **100**. The same applies to the connection portion between the wiring area **102** and the connector area **103**.

(35) The wiring pattern formed in the sensor area **101** extends into the wiring area **102**. The wiring pattern includes a signal line for transmitting signals output from the sensor parts mounted on the sensor area **101**, and a power line for supplying power to the sensor parts.

(36) In FIG. 2, at least a portion of the wiring area **102** is wound in a coiled form so as to form a

cylindrical outer periphery shape as a whole. As such, when the subdural sensor **10** is placed in vivo, the scalp can be easily and safely sutured around the wound portion using common suture techniques such as a purse-string suture. The wound portion may be coated with biocompatible materials, such as Parylene (registered trademark), silicone rubber, and the like. This makes it easier to maintain the wound shape.

(37) The connector area **103** is mounted with a connector **108** for connecting the subdural sensor **10** to external devices such as a control device. The connector **108** is connected to the wiring pattern extending from the wiring area **102**, transmits the signals output from the sensor parts mounted on the sensor area **101** to the external devices, and supplies the power from the external devices to the sensor parts. The connector **108** may be a wired connector (e.g., a male substrate-to-substrate connector) or a wireless connector. From the viewpoint of hygiene, a terminal part **10b** of the connector area **103** may preferably have a planar shape that tapers toward the end thereof so that it can be pulled outward from the inside of the scalp.

(38) Next, the sensor parts mounted on the sensor area **101** will be described. A blood flow measurement part using the principle of near-infrared spectroscopy (NIRS), a temperature measurement element (thermistor), an electrocorticogram (EcoG) measurement electrode, an intracranial pressure sensor, an acceleration sensor, a Doppler blood flow meter, or the like, may be mounted as the sensor part. FIGS. **1A** and **1B** show a plurality of blood flow measurement parts **110**, a plurality of temperature measurement elements **120**, and an intracranial pressure sensor **130**, as examples of the sensor parts. As described later, an electrocorticogram measurement electrode **114** (see FIG. **3**) is arranged at the bottom of the light reflection part **113** of the blood flow measurement part **110**. These sensor parts are mounted on the back **105** of the substrate **100**.

(39) It is preferable if biological information, such as cerebral blood flow, brain temperature, and electroencephalogram, can be acquired simultaneously at multiple locations in the brain. Therefore, in the present embodiment, three channels of each of the blood flow measurement part **110**, the temperature measurement element **120**, and the electrode **114** are provided. The number of channels of these sensor parts is not particularly limited, and the length of the sensor area **101** may be increased accordingly in order to increase the number of channels.

(40) Each sensor part, except for the light reflection part **113**, may preferably be coated with an insulation part **106** made of materials with biocompatibility and high light transparency, such as silicone rubber (e.g., polydimethylsiloxane (PDMS)). The insulation part **106** is placed in a round shape so as to wrap around the edges of each sensor part. The film **107** covers the entire surface of the substrate **100** and these insulation parts **106**.

(41) The blood flow measurement part **110** includes a light-emitting element **111** capable of emitting near-infrared light, a light-receiving element **112** capable of receiving the near-infrared light, and a light reflection part **113** arranged between the light-emitting element **111** and the light-receiving element **112**. The light-emitting element **111** emits near-infrared light into the brain. The light-receiving element **112** receives the near-infrared light reflected in the brain, and converts this near-infrared light signal into an electrical signal and outputs the same.

(42) FIG. **3** is a partial cross-sectional view showing the vicinity of the light reflection part **113**. The light reflection part **113** is formed of a metal with biocompatibility and good conductivity, such as platinum (Pt). The film **107** is formed such that, while it covers the boundary between the light reflection part **113** and the substrate **100**, the apex of the light reflection part **113** is exposed.

(43) As shown in FIG. **3**, the light reflection part **113** has a shape in which the circumferential part thereof bulges toward the inner periphery. Since the surface of the light reflection part **113** comes into contact with the brain surface, it is preferable for the light reflection part **113** to have a shape without edges, such as a dome shape.

(44) The light reflection part **113** reflects the near-infrared light again, which is emitted from the light-emitting element **111** and reflected in the brain, in the direction into the brain. As a result, the near-infrared light escaping to the outside of the brain may be reduced and the near-infrared light

reflected in the brain may be allowed to enter the light-receiving element **112** efficiently, therefore, the sensitivity for the gray matter portion located in relatively shallow regions within the brain can be improved. In addition, the light reflection part **113** reflects, at the circumferential part thereof, the near-infrared light which is emitted from the light-emitting element **111** in the direction into the brain. As a result, the near-infrared light may be suppressed from directly entering the light-receiving element **112** from the light-emitting element **111**, and the S/N ratio of the signal acquired in the blood flow measurement may be improved.

(45) At the bottom of the light reflection part **113**, an electrode **114** is arranged, which is mounted on the substrate **100** for electrocorticogram measurement, and the light reflection part **113** is placed so as to wrap this electrode **114**. In other words, the light reflection part **113** is electrically connected to the electrode **114** and also acts as an electrode for electrocorticogram measurement. The light reflection part **113** bulges from the back **105** of the substrate **100** and is easily brought into contact with the brain surface, making it possible to improve the detection sensitivity for the cortical potential. In addition, since the solder for mounting the electrode **114** is sealed by the light reflection part **113**, the electrical connection between the electrode **114** and the living body may be secured while safety to the living body is assured.

(46) The height of the light reflection part **113** is preferably approximately 0.5 mm or more, or more preferably approximately 0.7 mm or more, in order to fully wrap the electrode **114**. In addition, in order to smoothly insert the sensor area **101** under the dura mater, the height of the light reflection part **113** is preferably approximately 1 mm or less.

(47) (Usage Example of Subdural Sensor)

(48) FIG. 4 is a schematic diagram showing a usage example of the subdural sensor **10** according to one embodiment of the present invention. When using the subdural sensor **10**, an incision is made in the scalp **201**, a burr hole **203** is opened in the skull **202**, and a small incision is made in the dura mater **204**, from which the subdural sensor **10** is inserted into a subdural space **205** such that the surface **104** of the sensor area **101** is on the side of the dura mater **204**. Then, the subdural sensor **10** is pushed forward along the brain surface **206** (over the arachnoid mater) to be arranged at a target region of the brain surface **206**. On the other hand, in vitro, the connector **108** (see FIG. 2) provided in the connector area **103** is connected to a measurement system, and after checking the operation of the subdural sensor **10**, the scalp **201** is sutured around the wiring area **102**.

(49) Here, there are generally two ways to place the sensor under the dura mater: placing the sensor by exposing the brain after performing a craniotomy and making an incision in the dura mater; and, as described with reference to FIG. 4, making a small incision in the dura mater from a burr hole opened in the skull and inserting the sensor via the small incision in the dura mater along the brain surface into the subdural space. Since the incision is small, the latter way is advantageous over the former from the viewpoint of infection risks and stable retention of the sensor. It can be said that the latter way is also preferable in the case where the sensor is placed in the brain for a period of time (e.g., two weeks) in order to monitor biological (brain) information.

(50) On the other hand, when inserting the subdural sensor from the small incision in the dura mater, it is necessary to push the subdural sensor to the target region on the brain surface, and it is preferable to use materials that have flexibility, but have a certain degree of elasticity, as materials for the substrate, in order to perform the above task. Specifically, as mentioned above, a substrate may be used in which a resin material, such as polyimide, is coated with, for example, Parylene (registered trademark). However, with such subdural sensor, the operator must push the sensor forward through the subdural space such that the tip of the subdural sensor does not touch the brain surface, based on the sensation of the brain surface rather than visual confirmation. Therefore, advanced manipulation may be required for the operator in order to safely arrange the subdural sensor in the target region.

(51) Therefore, in the subdural sensor **10** according to the present embodiment, the planar shape of the tip part **10a** is curved convexly toward the outside, and the side shape of the tip part **10a** is

curved toward the surface **104**. As a result, the subdural sensor **10** can be safely and easily inserted into the subdural space **205** via the small incision in the dura mater **204**.

(52) In detail, since the tip part **10a** is curved toward the surface **104** side, the tip part **10a** is unlikely to come into contact with the brain surface **206** when the subdural sensor **10** is pushed forward along the brain surface **206** in the subdural space **205**. Therefore, this prevents the end of the subdural sensor **10** from touching the brain surface **206**. In addition, even if there is a living tissue in the same direction as the direction of travel of the subdural sensor **10**, since the subdural sensor **10** comes into contact with such living tissue at the underpart of the tip part **10a** (i.e., the back **105** of the curved tip part **10a**), the end thereof may still be suppressed from touching the brain surface **206**. Moreover, since the planar shape of the tip part **10a** is curved convexly and has no corners, the impact of the end of the sensor area **101** coming into contact with the surrounding living tissue may be mitigated.

(53) Further advantages of the subdural sensor **10** according to the present embodiment will be described below.

(54) In general, when a sensor is placed in vivo, the wiring is pulled out of the skull from the sensor arranged in the subdural space and then the scalp is sutured.

(55) In this regard, in the subdural sensor **10**, since at least a portion of the wiring area **102** of the substrate **100** is wound so as to form a cylindrical outer periphery shape as a whole, the scalp **201** can be safely sutured around the wound portion. The scalp **201** can also be easily sutured using versatile suture techniques such as a purse-string suture.

(56) In addition, the S/N ratio of the signal acquired in the blood flow measurement using near-infrared light can be improved in the subdural sensor **10** according to the present embodiment.

(57) Here, in the blood flow measurement using near-infrared light, a technique is also known, in which the near-infrared light escaping to the outside of the brain is reduced in order to improve the sensitivity for the gray matter portion by placing a reflective plate between the light-emitting element and the light-receiving element (see, for example, JP6296606 B).

(58) In contrast, in the present embodiment, the light reflection part **113** having a shape in which the circumferential part thereof bulges toward the inner periphery (e.g., a dome shape) is provided between the light-emitting element **111** and the light-receiving element **112**. As a result, the near-infrared light emitted from the light-emitting element **111** can be reflected in the direction into the brain and the near-infrared light escaping to the outside of the brain can be reduced. In addition, the light reflection part **113** allows for the near-infrared light emitted from the light-emitting element **111** to be suppressed from directly entering the light-receiving element **112**. In other words, the near-infrared light that enters the light-receiving element **112** without passing through the brain can be reduced.

(59) Accordingly, in the blood flow measurement using near-infrared light, the sensitivity for the gray matter portion can be improved and the S/N ratio of the acquired signal can also be improved.

(60) In addition, when the light reflection part **113** is formed of platinum, safety with respect to the living body may be improved and the electrical connection between the electrode **114** and the brain surface may be reliably secured, by placing the light reflection part **113** so as to wrap the electrode **114** for electrocorticogram measurement.

(61) Moreover, according to the present embodiment, both cerebral blood flow and electroencephalogram can be measured for a common region in the brain.

(62) (Variation of Subdural Sensor)

(63) FIG. 5A is a plan view schematically showing a first variation of the tip part **10a** of the substrate **100**. FIG. 5B is a side view schematically showing the same variation. The subdural sensor according to the present variation further includes a cover **140** covering the tip part **10a** of the substrate **100**. As shown in FIG. 5B, the cover **140** and the substrate **100** may be integrally coated with the film **107**, or the cover **140** may be placed on the substrate **100** coated with the film **107**.

(64) The cover **140** may preferably be formed of a flexible and biocompatible material such as silicone rubber. The cover **140** itself may have a shape that curves toward the surface **104** side. In this case, the tip part **10a** of the planar substrate **100** may be deformed by placing the cover **140** on the tip part **10a**. Alternatively, the cover **140** may be made to conform with the shape of the tip part **10a** by placing a flexible cover **140** on the tip part **10a** of the substrate **100** which is curved toward the surface **104** side.

(65) By providing such cover **140**, the end of the subdural sensor comes into contact with the living tissue in a gentle manner, further enhancing safety when inserting the subdural sensor into the subdural space. In addition, by providing the cover **140**, a load is applied to the tip part **10a**, so the lifting of the substrate **100** may be suppressed. As a result, various sensors may be allowed to come into close contact with the brain surface, and the accuracy of the acquired biological information may be improved.

(66) FIG. **6** is a plan view schematically showing a second variation of the tip part **10a** of the substrate **100**. The side shape of the present variation is the same as FIG. **5B**. As shown in FIG. **6**, safety when inserting the subdural sensor into the subdural space may also be enhanced by the cover **141** covering only the end of the tip part **10a** of the substrate **100**.

(67) In the case of placing a cover on the tip part of the substrate **100**, the tip part may not need to be curved toward the outer periphery side. This is because the end is brought into contact with the living tissue in a gentle manner by providing a cover (although this depends on the thickness and material of the cover) and safety can be improved compared to the case without a cover.

(68) FIG. **7** is a plan view schematically showing a first variation of the wiring area **102** of the substrate **100**. The subdural sensor according to the present variation further includes a tube **150** that accommodates therein at least a portion of the wiring area **102** of the substrate **100**.

(69) The tube **150** is formed of a flexible material such as silicone rubber. By accommodating at least a portion (specifically, a portion in the vicinity of the scalp when the subdural sensor is inserted into the subdural space) of the wiring area **102** in such tube **150**, the scalp can be easily and safely sutured around the tube **150**. The tube **150** can also provide the effect of protecting the wiring area **102**.

(70) FIG. **8** is a plan view schematically showing a second variation of the wiring area **102** of the substrate **100**. In the present variation, the wiring area **102** is wound around a core wire **151** in a coil form. The coil wire **151** is formed of a flexible material such as silicone rubber. The coil wire **151** and the wiring area **102** wound around the coil wire **151** may be coated in an integral manner with biocompatible materials, such as Parylene (registered trademark), silicone rubber, and the like. By using such core wire **151**, the strength can be increased while maintaining the flexibility of the wiring area **102**.

(71) FIGS. **9A** to **9C** are plan views each schematically showing a third variation of the wiring area **102** of the substrate **100**. Among which, FIG. **9A** shows the sensor area side, and FIGS. **9B** and **9C** show the connector area side. In the present variation, at least a portion of the wiring area **102** is divided into a plurality of strip-like areas along the longitudinal direction, and each of the plurality of strip-like areas is wound around the core wire **155**.

(72) The wiring area **102** is preferably divided into an area where a signal line pattern is formed (signal area **153**) and an area where a power line pattern is formed (power area **154**). In this case, the signal area **153** and the power area **154** are preferably wound in opposite directions to each other, and further, they are more preferably wound so that they are orthogonal to each other. In the present variation, again, the wound signal area **153** and power area **154** may be coated with biocompatible materials, such as Parylene (registered trademark), silicone rubber, and the like.

(73) In this way, by dividing the wiring area **102** into the signal area **153** and the power area **154** and winding them around the core wire **155**, the scalp can be easily and safely sutured around the wound portion. In addition, by winding the signal area **153** and the power area **154** in opposite directions to each other, it is also possible to avoid electromagnetic noise generated in the power

line being induced to the signal line and superimposed on the signal, thereby reducing the effect of noise on the signal.

(74) In this case, a connector area **156** on the signal area **153** side and a connector area **157** on the power area **154** side may be left divided as shown in FIG. **9B**, or they may be stuck together back-to-back as shown in FIG. **9C**.

(75) FIGS. **10A** to **10C** are diagrams each schematically showing an implementation of the intracranial pressure sensor **130**. Among which, FIG. **10A** is a plan view showing the surface **104** side of the substrate **100**, FIG. **10B** is a side view, and FIG. **10C** is a plan view showing the back **105** side of the substrate **100**. The intracranial pressure sensor **130** may be mounted in the vicinity of the tip part of the substrate **100**, and in this case, a reinforcement cover **142** surrounding the intracranial pressure sensor **130** may preferably be provided. In addition, the reinforcement cover **142** may be coated with the film **107** integrally with the substrate **100**.

(76) As shown in FIGS. **10A** and **10C**, the tip part of the reinforcement cover **142** has a shape that is curved convexly toward the outer periphery side. As shown in FIG. **10B**, the tip part of the reinforcement cover **142** is curved toward the surface **104** side. The reinforcement cover **142** is formed of flexible and biocompatible materials such as silicone rubber, and may have some weight in order to apply a load to the area in the vicinity of the intracranial pressure sensor **130**. As shown in FIG. **10C**, a U-shaped window **143** is formed on the back side of the reinforcement cover **142**, and the reinforcement cover **142** is fitted onto the substrate **100** so that the intracranial pressure sensor **130** is exposed at this window **143**. The intracranial pressure sensor **130** is coated with the insulation part **106**, similar to the above embodiment (see FIG. **1**).

(77) According to the present variation, since the intracranial pressure sensor **130** is arranged deep in the subdural space, and the reinforcement cover **142** suppresses the lifting of the intracranial pressure sensor **130**, the intracranial pressure can be measured more accurately. In addition, since the weight of the intracranial pressure sensor **130** and the reinforcement cover **142** also suppress the lifting of the substrate **100** as a whole, the accuracy of biological information acquired by other sensor parts may also be improved.

(78) FIG. **11** is a plan view schematically showing another implementation of the intracranial pressure sensor **130** and showing the back **105** side of the substrate **100**. On the back of the reinforcement cover **144** shown in FIG. **11**, a square-shaped window **145** that exposes the intracranial pressure sensor **130** is formed. In this way, by placing the reinforcing cover **144** so as to surround the intracranial pressure sensor **130**, a balanced load may be applied to the area where the intracranial pressure sensor **130** is arranged.

(79) FIG. **12** is a partial cross-sectional view showing a variation of the light reflection part. In the present variation, a temperature measurement element **120** is arranged inside the dome-shaped light reflection part **113**. An insulating material **115** such as polydimethylsiloxane (PDMS) is filled around the temperature measurement element **120**.

(80) By arranging the temperature measurement element **120** inside the light reflection part **113**, the temperature measurement element **120** may reliably be made to come into contact with the living tissue via the light reflection part **113**, and the heat conduction efficiency between the temperature measurement element **120** and the living tissue may be improved. Accordingly, the accuracy of the measurement of brain temperature may be improved. In addition, since centers of the respective regions to be measured regarding the electroencephalogram and brain temperature are aligned, cerebral blood flow, electroencephalogram, and brain temperature can be measured for a common region in the brain.

(81) FIG. **13** is a partial cross-sectional view showing another variation of the light reflection part. In the present variation, an electrode **116** for electrocorticogram detection is formed in a circular ring shape, and a light reflection part **117** is placed in a donut shape so as to wrap the electrode **116**. The temperature measurement element **120** is mounted on the substrate at the center of the light reflection part **117**.

(82) As shown in FIG. 13, when at least the circumferential part of the light reflection part **117** bulges toward the inner periphery, the near-infrared light may be suppressed from directly entering from the light-emitting element **111** into the light-receiving element **112** by the circumferential part reflecting such near-infrared light. Obviously, by reflecting the near-infrared light by the entire surface of the light reflection part **117**, the near-infrared light escaping to the outside of the brain may be reduced and the sensitivity for the gray matter portion may also be improved. Moreover, since centers of the respective regions to be measured regarding the electroencephalogram and brain temperature are aligned, cerebral blood flow, electroencephalogram, and brain temperature can be measured for a common region in the brain.

(83) Further, the shape of the light reflection part may be a shape where the circumferential part bulges in a straight line toward the inner periphery, such as a truncated cone, or the portion that comes into contact with the brain surface may be planar. In any case, the shape of the light reflection part may preferably be determined so that the edges thereof are not exposed.

(84) In the above-described embodiments and variations, the light-emitting element **111** and the light-receiving element **112** for blood flow measurement using near-infrared light are mounted on the back **105** side of the substrate **100**, but the light-emitting element **111** and the light-receiving element **112** may be mounted on the surface **104** side. In this case, the substrate **100** is preferably made of a material with high light transparency with respect to at least near-infrared light.

(85) FIG. 14 is a plan view schematically showing a variation of the substrate shape. As shown in FIG. 14, in the subdural sensor **10A**, the width in the short-length direction of the wiring area **102a** is smaller than the width in the short-length direction of the sensor area **101a**. The sensor area **101a** and the wiring area **102a** are smoothly connected in a tapered manner. In the present variation, the wiring area **102a** is not wound around, and the scalp is sutured around the planar wiring area **102a**.

(86) When the subdural sensor is pulled out from the suture point of the scalp after the sensor area of the subdural sensor has been placed in the subdural space for a predetermined period of time, a large force may be applied to the connection area between the sensor area and the wiring area.

(87) As such, as shown in FIG. 14, by forming the connection area M between the sensor area **101a** and the wiring area **102a** in a tapered form, the substrate is prevented from being applied with a force at a specific point thereof when the subdural sensor **10A** is pulled out, and the subdural sensor can be pulled out smoothly.

(88) As another variation, the widths of the sensor area and the wiring area in the short-length direction may be aligned. In this case, the subdural sensor can also be pulled out smoothly.

(89) FIG. 15 is a plan view schematically showing another variation of the substrate shape. FIG. 15 shows a plurality of sensor parts arranged in a line along the longitudinal direction of the substrate **100b** in the sensor area **101b** of the subdural sensor **10B**. In contrast to the maximum width W1 of these sensor parts in the short-length direction of the substrate **100b**, the width W2 in the short-length direction of the wiring area **102b** is equal to or greater than the width W1. In the present variation, the wiring area **102b** is not wound around, and the scalp is sutured around the planar wiring area **102b**.

(90) When the subdural sensor is pulled out from the suture point of the scalp after the sensor area of the subdural sensor has been placed in the subdural space for a predetermined period of time, a large force may be applied to the end of the sensor area which is wider than the wiring area.

(91) As such, as shown in FIG. 15, by setting the width W2 of the wiring area **102b** to be equal to or greater than the maximum width W1 of the sensor part, the deformation of the sensor area **101a** may be confined to only the end area where no sensor parts are arranged, when the subdural sensor **10B** is pulled out from the suture point of the scalp.

(92) FIG. 16 is a plan view illustrating the wiring in the connector area. As shown in FIG. 16, a signal line pattern **300** is formed in the wiring area of the subdural sensor **10C**, which transmits the signals output from each sensor part arranged in the sensor area **101**. Each signal line **301** included in the signal line pattern **300** is connected to a pin provided on the connector **108** mounted in the

connector area **103c**.

(93) A cable **400** shown in FIG. **16** is composed of, for example, a flexible substrate, and intermediates between external devices, such as control devices, and the subdural sensor **10C**. The cable **400** is mounted with an intermediary connector **410** (e.g., a female substrate-to-substrate connector), and a signal line pattern **420** corresponding to the signal line pattern **300** of the subdural sensor **10C** is formed thereon. In using the subdural sensor **10C**, the connector **108** is mated and electrically connected to the intermediary connector **410**. The signals output from each sensor part are input to the external devices, such as control devices, via the signal line pattern **300**, the connector **108**, the intermediary connector **410**, and the signal line pattern **420**.

(94) As shown in FIG. **16**, in the present variation, each signal line **301** included in the signal line pattern **300** is divided into two branches in the connector area **103b**, and the two branches are respectively connected to two different pins provided on the connector **108**. On the other hand, each signal line **421** included in the signal line pattern **420** on the cable **400** side is also divided into two branches in the vicinity of the intermediary connector **410**, and the two branches are respectively connected to two different pins provided on the intermediary connector **410**. In other words, the signal output from each sensor part of the subdural sensor **10C** is transmitted to the connector area **103c** by a single signal line **301**, passes through the two signal lines **301a**, **301b** branched off in the connector area **103c**, is transmitted to the signal lines **421a**, **421b** on the cable **400** side via the connector **108** and the intermediary connector **410**, and is then combined into a single signal line **421**.

(95) In this way, by transmitting a signal from the same origin through the signal lines **301a**, **301b** to the intermediary connector **410** via the two pins, a backup of the transmission route of the signals may be secured. In other words, even if a connection failure of one of the pins occurs between the connector **108** and the intermediary connector **410**, the signal can still be reliably transmitted to the external device via the other pin.

(96) The positions of the two pins connecting the branched signal lines **301a**, **301b** are not particularly limited, but it is preferable to select two pins respectively belonging to rows opposite to each other, as shown in FIG. **16**. As a result, even if the connector **108** and the intermediary connector **410** are connected in a slightly slanted manner, the connection at at least one of the pins may still be secured.

(97) FIG. **17** is a plan view showing a variation of the wiring pattern in the wiring area. In the present variation, a signal line pattern **510** of the wiring pattern is formed on one surface **511** of the substrate **100d** and a power line pattern **520** is formed on the other surface **521** of the substrate **100d** in the wiring area **102d**. Then, the signal line pattern **510** and the power line pattern **520** are each formed in a wave shape where peaks and troughs appear in an alternating manner, and also formed such that peaks and troughs in the signal line pattern **510** are staggered with respect to peaks and troughs in the power line pattern **520**.

(98) As a specific example of the wave shape, it may be a triangular wave (zigzag) shape, as shown in FIG. **17**, or a sine curve shape. Alternatively, the wave shape may have a shape where peaks and troughs of the triangular waves are curved, or a shape where arcs are connected. The distance between the peaks and troughs in each pattern may be adjusted appropriately according to the width in the short-length direction of the wiring area **102d** and the number of signal and power lines (i.e., the width of each pattern).

(99) In this way, by forming the signal line pattern **510** and the power line pattern **520** in an alternating manner so as to reduce the range in which these patterns run parallel, the effect of electromagnetic noise generated in the power line on the signals may be reduced.

(100) The signal line pattern **510** and the power line pattern **520** may also be formed on different layers on the same surface instead of forming them on the respective surfaces of the substrate **100d**. In short, each pattern should be formed so that, to the extent possible, the signal line pattern **510** and the power line pattern **520** do not run side-by-side.

(101) FIG. 18 is a partial cross-sectional view for describing an implementation of an element. When an element 531 contained in the sensor part is to be mounted on the substrate 100 by wire bonding, an insulation part 106 made of a flexible resin material, such as silicone, is preferably formed so as to cover the element 531 and wire 532 as a whole. As a result, the surrounding area of the element 531 can be reliably insulated, and the element 531 is brought into close contact with the substrate 100, further preventing the element 531 from falling off. In addition, by forming the film 107 on the upper layer, the flexibility of a mold 533 is regulated, and disconnection of the wire 532 is therefore prevented. As a variation, one element 531 may be bonded with two wires. In this case, the risk of disconnection may be reduced.

(102) FIG. 19 is a plan view schematically showing a further variation of the subdural sensor. The subdural sensor 10E shown in FIG. 19 is equipped with two sensor areas 101e and a wiring area 102e continuous with these sensor areas 101e. The configuration of the sensor parts provided in each sensor area 101e is the same as that shown in FIG. 1A. In this way, by branching the tip side of one substrate 100e to form a plurality of sensor areas, the measurable channel number of sensors can be increased and biological information about more regions can be acquired. In addition, by increasing the distance between the tips of the two sensor areas 101e, biological information on a wider region can be acquired.

(103) FIG. 20 is a plan view schematically showing a further variation of the subdural sensor. In the subdural sensor 10F shown in FIG. 20, a notch is formed at the tip side of the strip-like substrate 100f, and the substrate 100f is separated along the longitudinal direction so as to form two sensor areas 101f. A portion of the wiring area 102f of the substrate 100f is folded along the longitudinal direction, so that when the two sensor areas 101f are arranged on the same surface, the distance between the tip parts of the two sensor areas 101f is naturally increased. The configuration of the sensor parts provided in each sensor area 101f is the same as that shown in FIG. 1A. According to the present variation, the processing of the shape of the substrate 100f may be simplified.

(104) The present invention is not limited to the embodiments and variations described above, and may be carried out in various other forms within the scope that does not depart from the spirit of the present invention. For example, such various other forms may be formed by excluding some components from all of the components shown in the above-described embodiments and variations, or by appropriately combining the components shown in the above-described embodiments and variations.

(105) Further advantages and modifications may be easily conceived of by those skilled in the art. Accordingly, from a wider standpoint, the present invention is not limited to the particular details and representative embodiments described herein. Accordingly, various modifications can be made without departing from the spirit or scope of the general idea of the invention defined by the appended claims and equivalents thereof.

Claims

1. A subdural sensor that is to be arranged in a subdural space and acquires biological information about the brain, comprising: a substrate formed of a flexible material; and at least one type of sensor part mounted on the substrate, wherein the substrate has an elongated shape as a whole, wherein the substrate includes: a sensor area in which the at least one type of sensor part is mounted and a wiring pattern connected to the at least one type of sensor part is formed; a wiring area contiguous with the sensor area on one end thereof, the wiring pattern extending in the wiring area; and a connector area contiguous with the other end of the wiring area, the connector area being an area on which a connector to be connected to the wiring pattern extending from the wiring area is mounted, wherein a tip part of the sensor area has a planar shape that curves convexly toward an outer periphery, and a side shape that curves toward a first surface, the first surface being

on the side of a dura mater when the subdural sensor is inserted into the subdural space.

2. The subdural sensor according to claim 1, further comprising a cover that is formed of a soft material and covers the tip part of the sensor area.
3. The subdural sensor according to claim 1, wherein the at least one type of sensor part includes an intracranial pressure sensor mounted on an area in the vicinity of a tip part of the substrate.
4. The subdural sensor according to claim 3, further comprising a cover that is formed of a soft material, covers the tip part of the sensor area, and is arranged over a surrounding area of the intracranial pressure sensor.
5. The subdural sensor according to claim 1, wherein a width in the short-length direction of the wiring area is smaller than a width in the short-length direction of the sensor area, and a connection area between the sensor area and the wiring area is tapered.
6. The subdural sensor according to claim 1, wherein, in the sensor area, the at least one type of sensor part is arranged in a line along the longitudinal direction of the substrate, and wherein the width in the short-length direction of the wiring area is equal to or greater than the maximum width in the short-length direction of the substrate of the at least one type of sensor part arranged in the sensor area.
7. The subdural sensor according to claim 1, wherein the wiring pattern includes a signal line pattern and a power line pattern, wherein the signal line pattern is formed in a wave-shaped pattern on one surface of the wiring area, the wave-shaped pattern having peaks and troughs appearing in an alternating manner, and wherein the power line pattern is formed in a wave-shaped pattern on the other surface of the wiring area, the wave-shaped pattern having peaks and troughs that are staggered with respect to the peaks and the troughs of the signal line pattern.
8. The subdural sensor according to claim 1, wherein at least a portion of the wiring area is accommodated inside a tube formed of a flexible material.
9. The subdural sensor according to claim 1, wherein at least a portion of the wiring area is wound so as to form a cylindrical outer periphery shape as a whole.
10. The subdural sensor according to claim 9, wherein at least a portion of the wiring area is wound in a coiled form.
11. The subdural sensor according to claim 9, wherein at least a portion of the wiring area is divided into a plurality of strip-like areas along the longitudinal direction, and each of the plurality of strip-like areas is wound around one wire core.
12. The subdural sensor according to claim 11, wherein the wiring pattern includes a signal line pattern and a power line pattern, wherein at least a portion of the wiring area is divided, along the longitudinal direction, into a first strip-like area where the signal line pattern is formed and a second strip-like area where the power line pattern is formed, and wherein the first strip-like area and the second strip-like area are wound around the wire core in opposite directions to each other.
13. The subdural sensor according to claim 9, wherein an outer periphery surface of a wound portion of the wiring area is coated with a biocompatible material.
14. The subdural sensor according to claim 1, wherein the wiring pattern includes a signal line pattern that transmits a signal output from the at least one type of sensor part, and wherein each signal line included in the signal line pattern divides into two branches in the connector area and the two branches are respectively connected to two different pins provided on the connector.
15. The subdural sensor according to claim 1, wherein the at least one type of sensor part includes a blood flow measurement part that includes a light-emitting element capable of emitting near-infrared light, a light-receiving element capable of receiving near-infrared light, and a light reflection part arranged between the light-emitting element and the light-receiving element, and the light reflection part has a shape in which at least a circumferential part bulges toward an inner periphery.
16. The subdural sensor according to claim 15, wherein the light reflection part also serves as an electrode for electrocorticogram measurement.

17. The subdural sensor according to claim 15, wherein the at least one type of sensor part includes a temperature measurement element arranged inside the light reflection part.
 18. The subdural sensor according to claim 16, wherein the at least one type of sensor part includes a temperature measurement element arranged inside the light reflection part.
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